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Towards the understanding and prediction of climatic variations

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1. INTRODUCTION

One of the most marked and important developments in meteorology in the last few years has been the sudden upsurge of interest in the subject of climate and climatic change. There is an increasing awareness that climatic fluctuations, unpredictable in magnitude and duration, promise to have greater economic and social impact in the future than in the past, and some concern that man himself may induce inadvertent and irreversible changes. At the same time, improvements in paleoclimatic dating techniques have produced a much clearer record of the climatic history of the last one million years, while the availability of computers capable of executing more than 10 million instructions per second has encouraged the development of complex (though still over-simplified) numerical models for the simulation of global climates and for investigating the likely influence of various natural and man-made perturbations.

Although I think that the likelihood of major and potentially catastrophic changes in climate has been grossly exaggerated, the subject is of sufficient potential importance and concern to merit a sustained research programme aimed at determining past and current trends more reliably and at improving our understanding of the underlying mechanisms. Such a programme has been adopted as the second major objective of the Global Atmospheric Research Programme on which the Joint Organizing Committee (1975) has recently produced an excellent report.

In this lecture I shall attempt to review some of the problems of reconstructing and interpreting historical records, indicate to what extent the past may provide a guide to future changes, discuss some possible causes of climatic change, and describe the promise and limitations of numerical modelling for understanding and perhaps, eventually, predicting the evolution of the earth's climate.

Although the broad characteristics of the global climate have remained fairly stable and stationary since the end of the last glacial epoch some 10,000 years ago, climatological records based on the synthesis and statistical averaging of daily weather records of temperature, rainfall, etc., show considerable variations from year to year, from decade to decade, and from century to century. Although these fluctuations are generally small compared with the dramatic changes associated with an ice age, they may nevertheless have important economic and social consequences. Increasing world population and a desire to raise living standards have increased the pressure on the natural resources of food, water and energy, the balance between the supply and demand for which could be seriously affected by marginal changes in climate. With world grain reserves now only a few per cent of annual consumption, recent crop failures in the USSR and USA due to even temporary spells

TABLE 1. CHARACTERISTICS OF PALEOCLIMATIC DATA SOURCES

Proxy data source	Variable measured	Continuity of evidence	Potential geographical coverage	Period open to study (yr BP)	Minimum sampling interval (yr)	Usual dating accuracy (yr)	Climatic inference
Layered ice cores	Oxygen isotope concentration, thickness (short cores)	Continuous	Antarctica, Greenland	10,000	1-10	±1-100	Temperature, accumulation
	Oxygen isotope concentration (long cores)	Continuous	Antarctica, Greenland	100,000+	Variable	Variable	Temperature
Tree rings	Ring-width anomaly, density, isotopic composition	Continuous	Midlatitude and high-latitude continents	1,000 (common) 8,000 (rare)	1	±1	Temperature, runoff, precipitation, soil moisture
Fossil pollen	Pollen-type concentration (varved core)	Continuous	Midlatitude continents	12,000	1-10	±10	Temperature, precipitation, soil moisture
	Pollen-type concentration (normal core)	Continuous	50°S to 70°N	12,000 (common) 200,000 (rare)	200	±5%	Temperature, precipitation, soil moisture
Mountain glaciers	Terminal positions	Episodic	45°S to 70°N	40,000	—	±5%	Extent of mountain glaciers
Ice sheets	Terminal positions	Episodic	Midlatitude to high latitudes	25,000 (common) 1,000,000 (rare)	—	Variable	Area of ice sheets
Ancient soils	Soil type	Episodic	Lower and mid-latitudes	1,000,000	200	±5%	Temperature, precipitation, drainage
Closed-basin lakes	Lake level	Episodic	Midlatitudes	50,000	1-100 (variable)	±5%	Evaporation, runoff, precipitation, temperature
Lake sediments	Varve thickness	Continuous	Midlatitudes	5,000	1	±1	Temperature, precipitation
Ocean sediments (common deep-sea cores, 2-5 cm/1000 yr)	Ash and sand accumulation rates	Continuous	Global ocean (outside red clay areas)	200,000	500+	±5%	Wind direction
	Fossil plankton composition	Continuous	Global ocean (outside red clay areas)	200,000	500+	±5%	Sea-surface temperature, surface salinity, sea-ice extent
	Isotopic composition of planktonic fossils; benthic fossils; mineralogic composition	Continuous	Global ocean (above CaCO ₃ compensation level)	200,000	500+	±5%	Surface temperature, global ice volume; bottom temperature and bottom water flux; bottom water chemistry
(rare cores > 10 cm/1000 yr)	As above	Continuous	Along continental margins	10,000+	20	±5%	As above
(cores < 2 cm/1000 yr)	As above	Continuous	Global ocean	1,000,000+	1000+	±5%	As above
Marine shorelines	Coastal features, reef growth	Episodic	Stable coasts, oceanic islands	400,000	—	±5%	Sea level, ice volume

of bad weather point to the difficulties that could arise if there were a widespread and persistent deterioration in the climate of the major food-producing areas. Unfortunately our understanding of the mechanisms and causes of climatic trends and fluctuations is inadequate to allow of their prediction. It is not even clear whether they are brought about by internal changes in the atmospheric-oceanic system or by changes in external factors such as the sun's radiation. The problem may, in the foreseeable future, be complicated further by the possibility that human activities through the production of additional carbon-dioxide, aerosols and waste heat may also cause significant changes in weather and climate.

2. A BRIEF SURVEY OF PAST CLIMATES

An essential prerequisite to understanding and evaluating climatic changes is a coherent and consistent reconstruction of the earth's climatic history, our knowledge of which, beyond the short instrumental record of only 200–300 years, rests very largely on ingenious and painstaking analyses of tree-rings, fossil flora and fauna in deep-ocean sediments, fossil pollen in layered soils, isotopic composition in ice cores from polar ice-caps, sea-level changes, etc. In view of the limitations of these techniques, see Table 1, and the fact that such data are available for only a rather few sites, it is remarkable that so coherent a picture of past climates has emerged, although there are many gaps to be filled.

At present we are in an interglacial epoch within an ice age which has existed for at least two million years. The present ice age is the most recent in a series of at least three which, at intervals of about 300 million years, have interrupted otherwise considerably

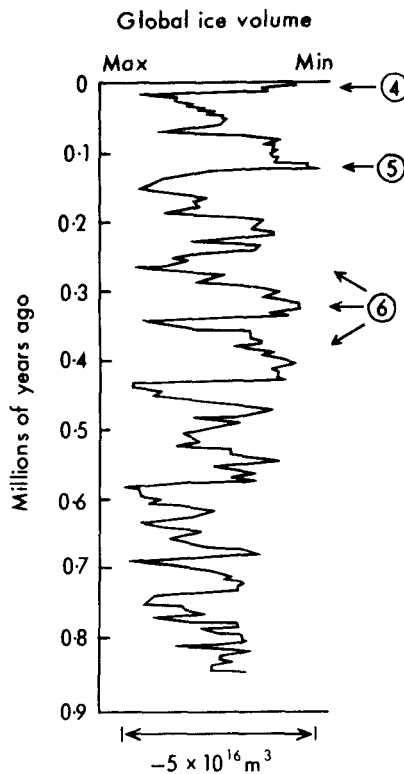


Figure 1. Fluctuations in global ice volume (or sea level) during the last million years as recorded by changes in the isotopic composition of fossil plankton in a deep-sea core from the equatorial Pacific. (Shackleton and Opdyke 1973.)

warmer and ice-free conditions on earth. The origin of the present ice age is not known for certain but may well have coincided with the movement of the Antarctic continent towards its present polar position about 50 million years ago and the gradual isolation of the Arctic ocean from the mid-latitude oceans by the North America–Greenland–Euroasia land mass. Evidence from ocean-bottom sediments suggests that the Arctic ocean has been largely ice-covered at all times during the last million years.

The chronology of climates existing more than a million years ago is poorly documented and even that of the last million years, of the Pleistocene, has been firmly established only in the last few years, mainly by the isotopic analysis of fossil plankton in deep-ocean sediments. This record, see Fig. 1, shows eight alternations between cold glacial epochs and relatively warm interglacial epochs occurring at intervals of order 100,000 yr. These transitions were accompanied by large fluctuations in the northern hemisphere ice sheets and sea-level variations of order 100 m. The warmest stages of the interglacial periods, such as we are experiencing at present, have lasted for only about 10,000 yr and the transitions from glacial to interglacial have usually been more abrupt (~ 2000 yr) than those from interglacial to glacial (5–10,000 yr). There is, however, some recent evidence to suggest that the *initial* phase of the transition to a glacial epoch may be very short, perhaps only some hundreds of years. The transition from the warmest stages of an interglacial to a fully developed glacial state is associated with a fall in average global temperatures of about 10 degC and a three-fold increase in the total volume of ice, but the last cold epoch was interrupted by five marked fluctuations of amplitude of 3–5 degC giving warmer and colder stages (interstadials) at intervals of about 20,000 yr – see Fig. 2. In addition the records show fluctuations on time-scales ranging from 20,000 to < 100 yr, some of which are of remarkably large amplitude and short duration.

During the past 20,000 yr some of the changes were as severe as any that occurred during the last million years. At the last glacial maximum, about 18,000 yr ago, extensive areas of the northern hemisphere were covered with continental ice sheets, the sea level having dropped by about 120 m and the sea-surface temperatures in the North Atlantic by

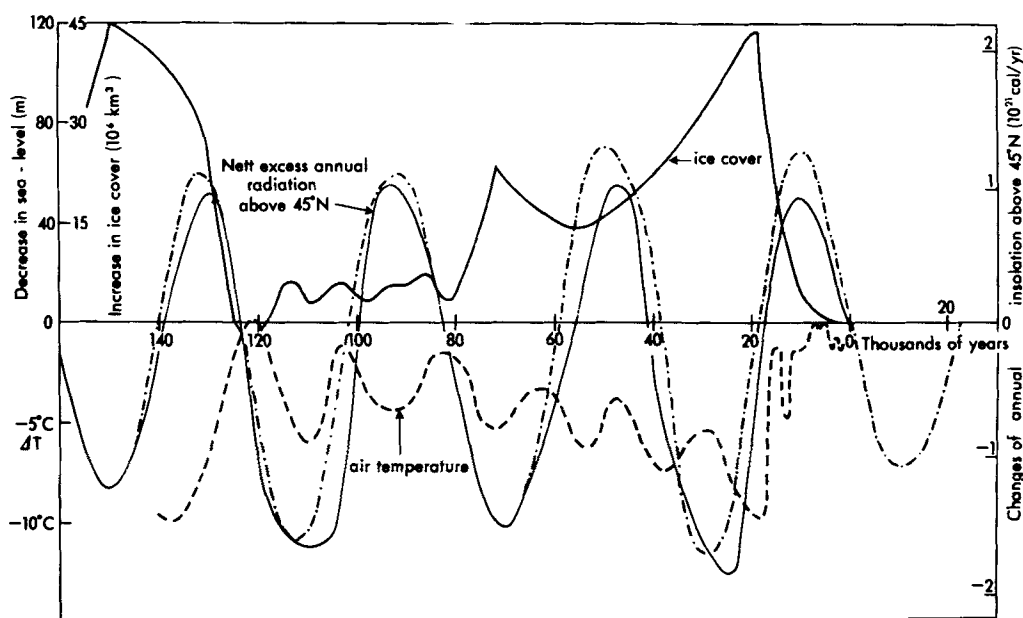


Figure 2. Changes in global ice cover, northern hemisphere air temperatures, and total insolation north of 45°N over the last 150,000 yr.

as much as 10 degC since the climatic optimum in 120,000 BP (Before Present). During the recession of the North American ice sheet, which took about 8000 yr between 14,500 and 6500 BP, a sequence of five alternate warm and cold intervals with amplitudes of up to 6 degC occurred within less than 2000 yr over Greenland and much of North America and left their mark even in the Mediterranean. The last interglacial maximum was reached about 6000 BP and climates since then have undergone a gradual cooling. This trend was interrupted by three shorter periods of more rapid cooling by 1–2 degC, each followed by a temperature recovery, the last of which occurred about 2400 BP.

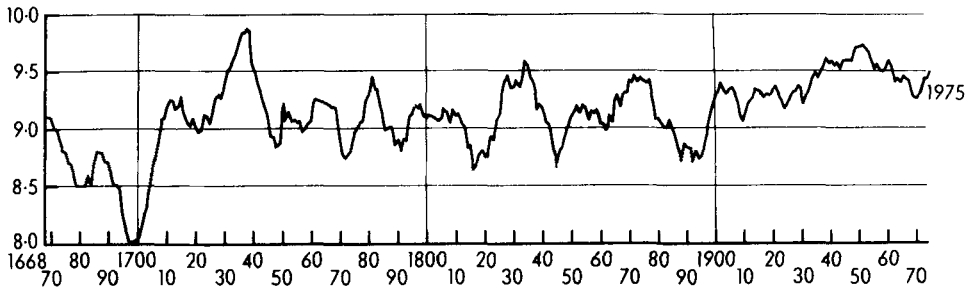


Figure 3. Ten-year running means of Central England temperatures from 1650 to 1960 as compiled by Manley and now extended to 1975.

Prominent among the temperature fluctuations of the last 1000 years are the warm period of the Middle Ages (1150–1350 AD) followed by the so-called ‘Little Ice Age’ of 1500–1850 AD and thereafter (see Figs. 3, 4), by an upward trend of global temperatures beginning about 1880 and continuing until 1940. Between 1940 and 1970 the mean temperatures drifted downwards by about 0.3 degC, half-way back to the 1880 levels; but this cooling trend, which was confined largely to latitudes above 50°N, has been arrested during the last five years, especially in the Arctic basin (see page 483).

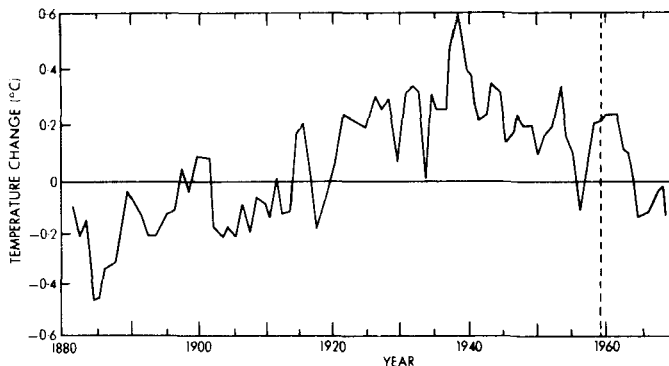


Figure 4. Recorded changes in the annual mean temperature of the northern hemisphere since 1880. (After Budyko 1969.)

3. THE ANALYSIS, INTERPRETATION AND PREDICTIVE VALUE OF CLIMATOLOGICAL RECORDS

In analysing, interpreting, extrapolating, and making deductions from historical records, it is important to realize that the magnitudes of climatic averages themselves may vary not only with the location but with the period and length of the record. Unfortunately records are often too sparse and incomplete to provide continuity and coherence in space and time. Since climates are not stationary, a particular period of, say, 50 yr, may not be

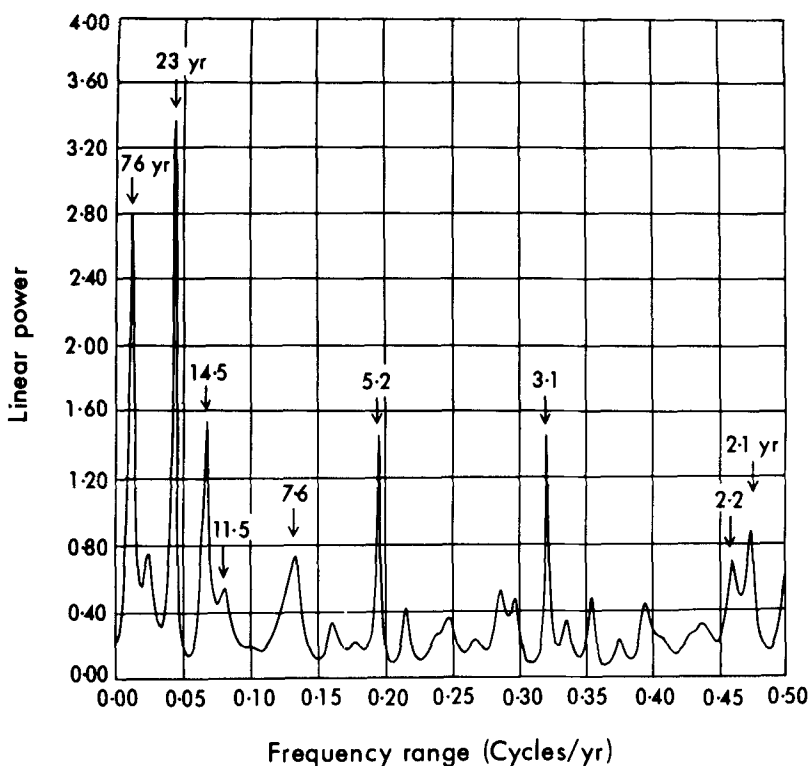


Figure 5. Power spectrum of the Manley record of Central England temperatures with the long-term trend removed.

representative or provide good guidance as to what may be expected in the next 50 yr; indeed the period 1931–1960, on the statistics of which many recent investigations have been based, was probably one of the most abnormal 30-yr periods in the last thousand years. Furthermore, statistical averages often conceal more than they reveal and an adequate description of the climatic state should include at least the variance and the extreme values, and a correlation between two or more quantities should be accompanied by the results of specified significance tests.

The results of an analysis of a climatological record, e.g. a power spectrum or an harmonic analysis, may be influenced by the degree to which the data have been pre-conditioned by smoothing, filtering, or the removal of long-term trends. For example, Fig. 5 shows a power spectrum, derived by the maximum entropy method, of Manley's 250-yr (1700–1950) record of annual mean Central England temperatures from which the linear long-term trend containing about 20% of the variance has been removed. The taller the peak for a given area enclosed beneath it, the greater its statistical significance. Fig. 5 shows significant peaks at 2.1, 5.2, 7.6, 14.5, 23 and 76 yr. The first peak, which may be identified with the quasi-biennial oscillation (QBO), contains about 10% of the total variance and is significant at the 5% level according to Snedecor's F test. Some of the lower frequency peaks are probably the result of non-linear interactions between higher-frequency periodicities but the 23-yr peak, containing about 8% of the variance and significant at the 0.1% level, may be associated with the double sunspot cycle. However, the small 11.5 yr peak contains <4% of the variance and is barely significant at the 10% level.

Harmonic analysis of the same record also reveals the QBO, in which the summers in odd years tend to be warmer than those in even years, and a marked 23-yr cycle, but again

an 11.5yr cycle is barely detectable. The most marked feature of the record is the rising trend from about 1880 to 1940. The annual rainfall totals for England and Wales, when harmonically analysed, show a weak peak at 2.4yr and even weaker peaks at 4, 5, 6 and about 50yr. The summer rainfall shows no definite trend over the 250yr of the record but the winter rainfall shows a clear upward trend in phase with the rising temperatures from 1850 to 1940.

The dangers of drawing uncritical, facile conclusions from apparent correlations between the march of two parameters over only a very limited period are illustrated in Fig. 6. King (1973, 1974) published some apparently striking relationships between the yields of various crops and the sunspot cycle using 5-yr running means over periods of only 25yr. Analyses of the data carried out in the Meteorological Office by Folland (un-

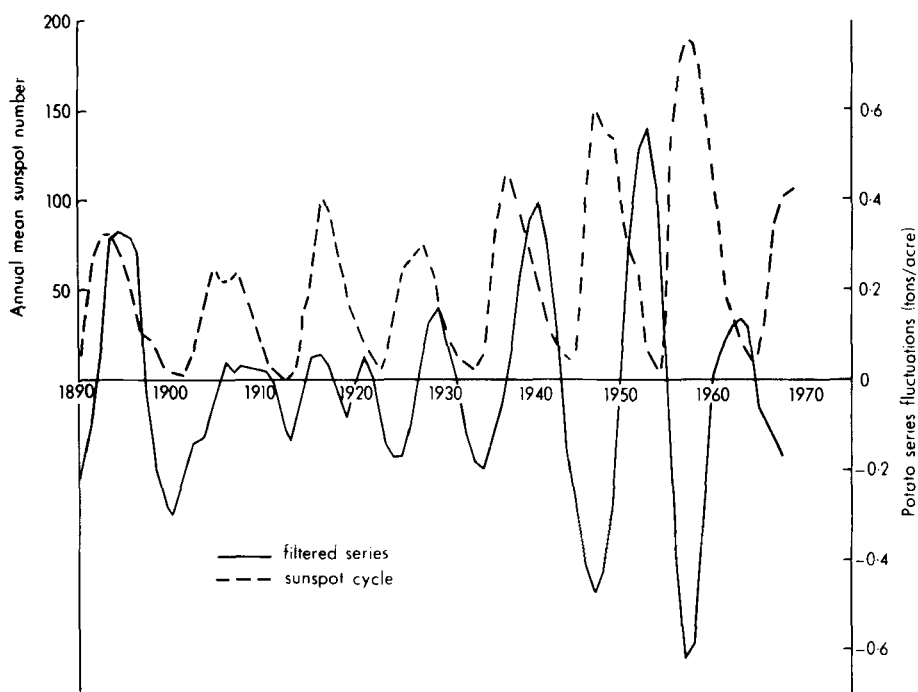


Figure 6. Correlations for the period 1890–1970 between the sunspot cycle and the series of potato yields in Gt. Britain filtered to pass cycles of 7 to 15yr.

published) show that all the crop-yield series have strong highly non-linear trends due to improved methods of production and, when these are removed, the quasi-11-yr peaks in the power spectra contain only a small proportion of the variance in all cases. Moreover, an analysis of the peaks using quasi-orthogonal filters shows generally poor correlations of the filtered series with the sunspot series, with frequent relative changes of amplitude and phase between the two. This is illustrated in Fig. 6, where the anti-phase relationship between the sunspot cycle and potato yields cited by King for the period 1935–1959 obviously did not hold over the longer period 1890–1935.

A great deal of such effort has been devoted to searching the historical records for cycles, regular patterns or correlations that may have some prognostic value. In order to have any predictive value, the records must contain either:

(a) a well-established trend the cause of which is sufficiently understood to indicate that it is unlikely to change over the forecast period; or

- (b) a well-defined periodicity established over several regular cycles; or
- (c) variations that are demonstrably controlled by some known periodic forcing factor.

In fact, very few records satisfy any of these conditions.

Analysis of paleoclimatological records for the last million years, in particular the power spectrum of the isotopic composition of fossil plankton in deep-ocean cores (which is linked with fluctuations in sea-level and total global volume of ice) reveals marked peaks

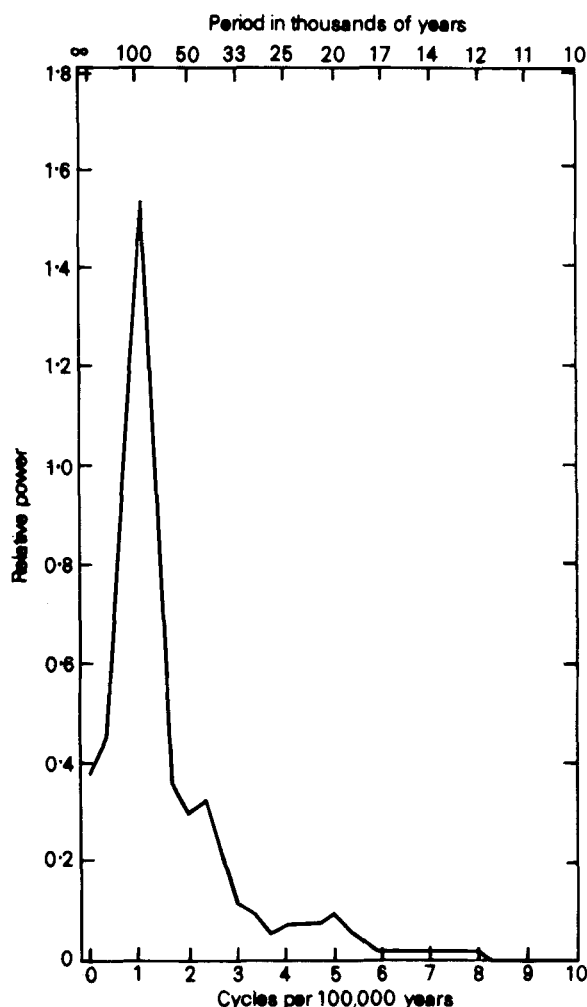


Figure 7. Power spectrum of a time series of observations of the oxygen-isotope content of fossil plankton in a deep-sea core from the equatorial Pacific which indicates fluctuations in global ice volume over the last 600,000 yr. (From the US National Academy Report, 'Understanding Climatic Change', 1975, p. 144.)

at periods of about 100,000, 40,000 and 20,000 yr – see Fig. 7. These have been associated with secular variations in the earth's orbit about the sun, namely changes in the eccentricity of the orbit (period 96,600 yr), in the obliquity of its axis (40,000 yr) and the precession of the longitude of the perihelion (21,000 yr) as first suggested by Milankovitch (1930). There are also two much smaller and ill-defined peaks at about 2500 yr and 100–400 yr which appear to be significant above the random noise but neither has a known cause or been related to any external factor that might exert a controlling influence on the climate. The

isotopic composition of a core from the Greenland icecap by Dansgaard *et al.* (1971), claimed to provide a record of air temperature over the last 10,000 years, show peaks in the last 1000 years at 77- and 177-yr intervals. However, although this record has been adjusted to match some of the longer-period climatic variations deduced from deep-sea sediments, it shows rather poor agreement with Manley's 250-year actual temperature record for Central England (see Fig. 8) and this casts some doubt on the reliability of the ice-core dating when applied to the rather small fluctuations of the last few centuries.

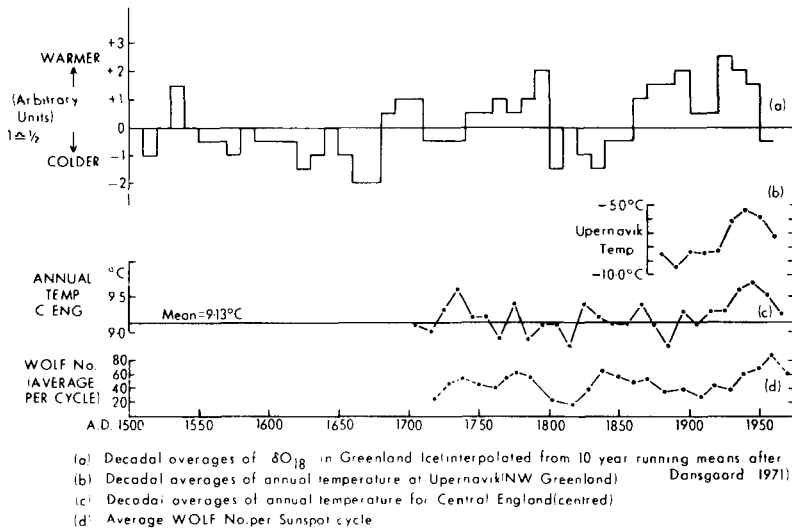


Figure 8. Historical temperature records as described in the legend.

Apart from the very long-term 'astronomically-induced' Milankovitch changes, the records are so dominated by irregular fluctuations, with few, if any, regular cycles or well-established trends, they are of little value for predicting future events. The unrewarding search for repeated patterns or cycles is well summarized in a recent report by the US National Academy of Sciences (1975) as follows:

'Since the development of modern techniques of time-series analysis, in particular those involving the determination of the variance (or power) spectrum, it has become clear that almost all the alleged climatic cycles are either (1) artifacts of statistical sampling, (2) associated with such small fractions of the total variance that they are virtually useless for prediction purposes, or (3) a combination of both.'

Nevertheless, the records can be used to estimate, within broad limits, the probability that fluctuations of a given magnitude and duration may occur within a specified time interval and to judge whether a particular event is 'abnormal' in the sense that it lies well outside expectation based on the variance of the long-term record.

In this light we may point out that recent disasters such as the droughts in the Sahel region of Africa, far from being unprecedented, have parallels in the past – as shown in Figs. 9 and 10. Although wet and dry spells tend to occur in anomalous groups of years, they do not occur at regular predictable intervals. The recent droughts have been attributed to a change in the global atmospheric circulation leading to a southward shift of the major climatic zones, including the Sahara desert: but a recent careful analysis by Painting in the Meteorological Office shows that the situation is not as simple as that and that, during the last five years, the sub-tropical high-pressure belt has actually moved slightly northwards.

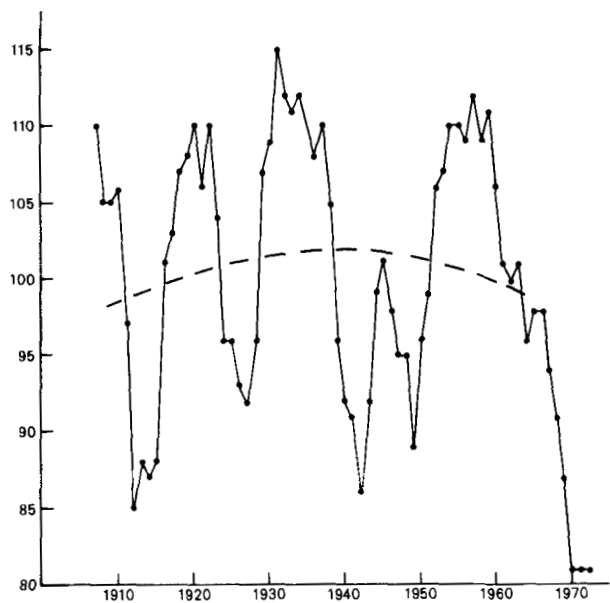


Figure 9. Five-year means of annual rainfall in the Sahel region of West Africa expressed as percentages of the 1905-1970 average. The dashed curve shows the long-term trend.

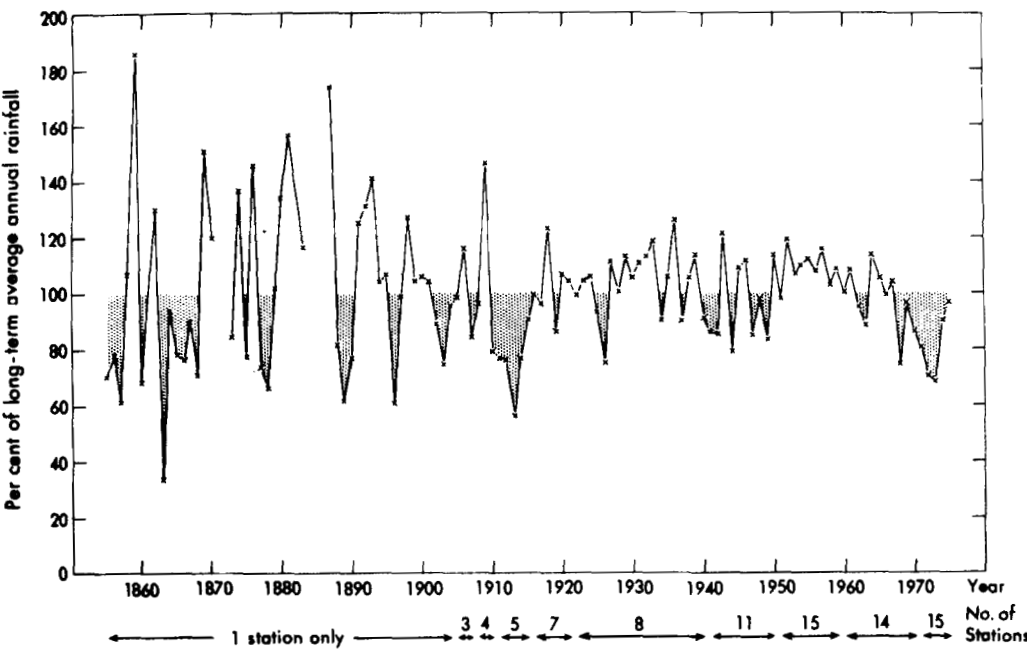


Figure 10. Annual rainfall in the Sahel region of Africa expressed as a percentage of the long-term average and showing anomalous groups of wet and dry years.

There is, I believe, no real evidence to indicate that the global pattern of climate is undergoing a rapid and permanent change or that recent events are abnormal in that they lie outside expectation based on the variance of the historical record. A more realistic, if less dramatic, assessment would be that fluctuations of climate will continue to occur with about the same magnitude, frequency and variability as in recent centuries, superimposed on

longer-term trends, the onset and reversal of which cannot yet be predicted in advance. At present we are in a warm interglacial period, the duration of these in the past have averaged about 50,000 yr. It is probable that the present very warm interval, which has already lasted for about 10,000 yr, will eventually give way to a period of colder climate. Statistically the chances that such a transition will begin in the next hundred years may be placed at about 1 in 100 but the full drop of 10 degC or so would probably be spread over several thousand years. There is a rather higher probability that a cooling may set in but not be carried through to the full glacial conditions. The chances of a prolonged cold, but far from glacial, spell within the next century, with average temperatures lower by about 1 degC, such as occurred between 1500 and 1850, must be put quite high, about 1 in 5. However, there is no physical basis for predicting either the timing or magnitude of such changes because we do not yet understand the underlying causes. Likewise there is no real basis for the alarmist predictions of an imminent ice age which have been largely based on extrapolation of the 30-yr trend of falling temperatures in the northern hemisphere between 1940 and 1965 (see Fig. 4). Apart from the strong dubiety of making a forecast from such a highly fluctuating record by extrapolation of such a short-period trend, there is now evidence that the latter has been arrested. A recent investigation in the Meteorological Office by Painting (1976) shows that since 1970 there has been a rise of about 0.2 degC in both the surface and 1000–500 mb temperatures in the Arctic basin, a marked reduction in both the land snow cover (except for an area north of the Himalayas) and in sea ice, and stronger westerly winds with less frequent outbreaks of northerlies leading to a succession of mild winters in Europe. In middle latitudes the cooling trend of the 1960s has continued but at a reduced rate.

4. SOME POSSIBLE CAUSES OF CLIMATIC CHANGE

(a) *Changes in solar output*

In seeking possible causes of major climatic fluctuations one may contemplate changes in the external forces or agencies acting on the atmosphere, internal changes within the atmosphere–ocean system itself or, more likely, an interactive combination of both.

Since the sun is the primary source of energy for driving the global atmospheric circulation, it is natural to consider possible variations in either the sun's output or, at any rate, secular variations in the intensity of the solar radiation reaching the earth, as likely mechanisms of exerting periodic control on the climate. However, the quest for the necessary evidence and understanding of such sun-weather relationships has proved very difficult and largely unrewarding.

Much effort has been devoted to monitoring the intensity of the total solar radiation (the so-called 'solar constant') and to examining the record for evidence of cyclic or periodic changes. Since most of the measurements have been made at the earth's surface or within its atmosphere, it has proved difficult to determine whether small variations in the measured quantities are caused by changes in the solar output or by fluctuations in atmospheric attenuation. Although these problems can, in principle, be overcome by making measurements from a satellite, space-borne radiometers are not yet capable of detecting variations of order 0.1%. At present there is no convincing evidence that the total annual solar flux incident on the top of the atmosphere varies by as much as 1%, even during solar disturbances. During solar flares and other periods of increased solar activity there is a marked increase in the emission of ultraviolet and corpuscular radiation but these contribute only about 0.1% to the total radiative flux.

A simple radiative heat balance calculation indicates that a 1% change in the solar constant would cause the mean equilibrium temperature of the earth's surface, with an

average albedo of 0.3, to change by 0.6 degC. A more detailed computation by Wetherald and Manabe (1975), based on a simplified dynamical model of the global circulation with fixed cloudiness and in which the ocean is treated as a wet surface for evaporation but has no thermal capacity and transports no heat, indicates that a 2% increase in the solar constant would produce a rise of 3 degC in the mean global surface temperature, but a decrease of 2% would produce an average temperature drop of 4.3 degC. The induced changes are calculated to be much greater near the poles than at the equator because of the marked changes in snow cover and in albedo. The most marked effect was upon the precipitation where a 6% change (from -4% to +2%) in the solar constant produced a 27% increase in the area-mean rates of precipitation. There was a disproportionately greater *decrease* in the rate of snowfall because the poleward retreat of the zone of sub-freezing surface temperatures shifted the latitude of maximum snowfall away from the highly baroclinic zone of high total precipitation in middle latitudes. Because of the limitations and simplifications of the model the actual numerical magnitudes of the computed changes cannot be regarded as reliable within a factor of two. This particular model, which allows the ocean-surface temperatures to change in accordance with the local radiation balance, with no heat storage or transport in the oceans, and fixed cloudiness, almost certainly exaggerates the changes in the mean surface temperature and precipitation. Even so, the implications are that a 1% change in solar constant would change the surface temperature by only 1–2 degC and the rainfall by <5%. Since, as we have seen, fluctuations in the solar constant in recent centuries have probably been well below 1%, it is not surprising that even the high-resolution instrumental and tree-ring records fail to show *marked* periodic fluctuations associated with the sunspot cycle. While diligent searches of the vast mass of available data, with thousands of stations to choose from, may well turn up striking correlations between weather parameters and the sunspot cycle (or even less plausibly related events) for a small area over a limited period, few such correlations turn out to be extensive in both space and time and to persist without changes in magnitude and phase: there is little reason to infer that the phenomena are causally connected.

The nature of the problem may be illustrated by the fact that a recent computer analysis of northern hemisphere surface pressure data carried out in the Meteorological Office failed to confirm the existence of a marked 22-year (double sunspot) cycle claimed by Maksimov and Slepcev-Sevlevic (1971) and showed that any slight variation that there may have been within an 11-year cycle could not easily be separated from the larger year-to-year variations which arise from other causes. Analysis of the annual mean Central England temperature record for 1750–1974 showed that while the ‘odd’ epochs of the Hale 22-yr (double) sunspot cycle were warmer than the even epochs by a few tenths of a degree centigrade between the mid-18th and the mid-20th centuries, previous to that time, and since, a change of phase seems to have taken place leaving the ‘odd’ half of the Hale cycle the colder of the two. On the other hand, findings such as that of Jenkinson (unpublished) that the winter (but not the summer) rainfall data for the United Kingdom since 1873 shows a peak of 7½% above average just before the sunspot maximum and a 7½% deficiency just before the minimum cannot be completely dismissed and sustain interest in the subject even though they have no convincing explanation.

Claims have also been made for the existence of longer periodicities of 80–90 and 180–200 years in climatic records, especially of air temperature. The presence of a significant peak at 76 yr in Manley’s record of Central England temperatures (Fig. 5) and of peaks at 77 and 177 yr in the oxygen isotopic record of the Greenland ice core have already been noted. There has been a tendency to associate these periodicities with similar oscillations in solar activity, but the power spectrum of the mean annual sunspot number peaks at 90 yr and the recent part of the Greenland ice core record correlates poorly with measured

temperatures and with the record of sunspot numbers. One is therefore inclined to agree with Miles (1974) that 'the evidence for periodic oscillations of 80–90 and 180–200 years in solar activity does not seem strong enough to support the slender evidence for cycles of this length in atmospheric temperature or circulation patterns and allow the conclusion that the latter are modulated by periodic variations in the solar constant'.

(b) *Long-term changes in solar insolation*

On much longer time scales the intensity of the solar radiation incident on the atmosphere varies due to secular changes in the earth's orbit about the sun, namely changes in the eccentricity of its orbit with a periodicity of 96,600 yr, changes in the obliquity of its axis (40,000 yr), and changes due to the precession of the longitude of the perihelion equivalent to precession of the solstices and equinoxes (21,000 yr). The consequential variations

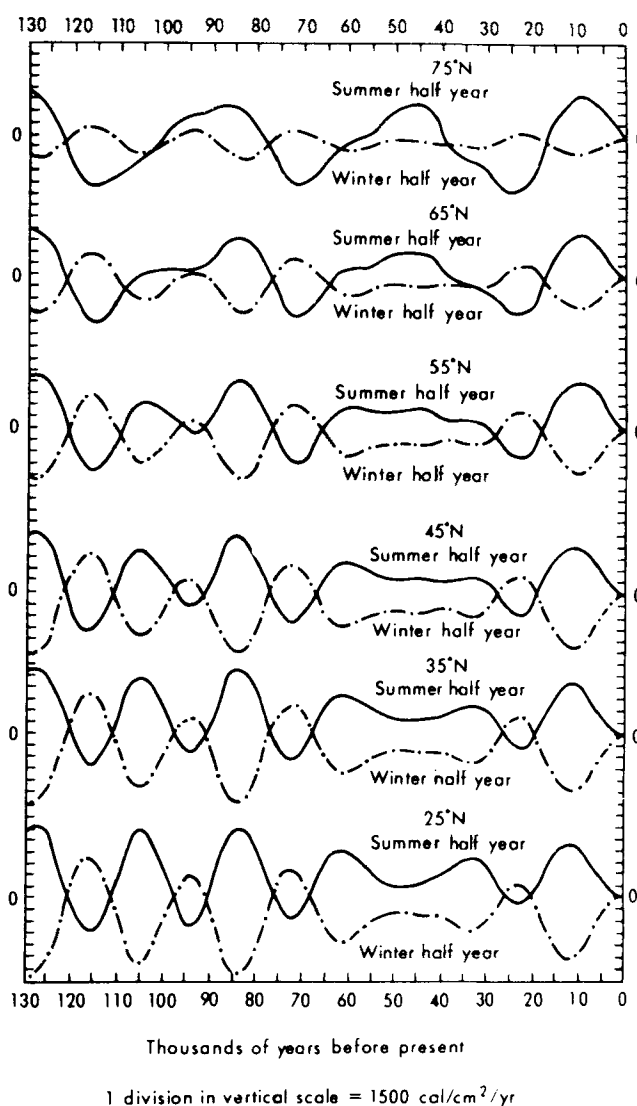


Figure 11. The variations over the last 130,000 yr of the radiative flux at latitudes between 25° and 75°N in the summer and winter half years according to Milankovitch (1930).

in the incident solar radiation were calculated, as a function of season and latitude, for the last million years by Milankovitch (1930, 1938). The variations over the last 130,000 yr in the northern hemisphere are plotted in Fig. 11. In high latitudes they are dominated by the 40,000 yr cycle in which the obliquity of the earth's axis last reached a maximum value and produced maximum insolation at high latitudes 10,000 years ago. This coincided with the rapid termination of the last glacial epoch as indicated by melting of the ice sheets and a rapidly rising sea-level. At low latitudes the 20,000 yr cycle dominates.

The fluctuations in Fig. 11 are much larger than any observed secular fluctuations in the 'solar constant' and range from about +1% at 10,000 BP to -2% at 25,000 BP in the annual incident solar radiation at 65° lat. The summer and winter components show larger fluctuations which tend to cancel in the annual mean values. For example, at 10,000 BP the radiation received in the summer half year was 4% greater at all latitudes than at present and Milankovitch estimated that the surface temperature exceeded present values by 4 degC at 65°N and by nearly 5 degC at 25°N whereas the change in the annual mean temperature at 65°N was only 0.7 degC. Even this latter figure has been challenged as being too high, for example by Sawyer (1966) who arrived at a value of 1/6 deg C. It is therefore of interest to compare Milankovitch's estimated temperature changes with those predicted by numerical models.

Shaw and Donn (1968), using Adem's simple energy-balance model, calculated, in close agreement with Milankovitch, that when the radiation deficiency was a maximum (2%) at 25,000 BP, the annual mean temperature at 65°N should have been 1.4 degC lower than at present. Wetherald and Manabe (1975), using the simplified global circulation model (GCM) described above, calculated that a 2% decrease in 'solar constant' would lower annual surface temperatures at 65°N by 5 degC and global mean temperatures by 4.3 degC. A 4% decrease in solar constant was computed to cause a drop of 10 degC in global mean temperatures without, however, providing an unstable ice cap spreading all the way to the equator. As we have argued on p. 484, these particular model results probably overestimate the temperature changes but, even so, suggest that Milankovitch's estimates of the temperature fluctuations induced by changes in the earth's orbit are not greatly exaggerated. Moreover, since the summer radiation is of greater importance than the annual mean values in determining the accumulation of ice and snow from year to year, particularly around 65°N where the earth is ringed by mountains with high winter snowfall, then it seems that the Milankovitch variations in insolation, producing temperature changes of 4-5 degC, could well be responsible for initiating the advance and retreat of the ice sheets.

5. AN EXTENSION OF THE MILANKOVITCH THEORY OF ICE AGES

Fig. 2 shows how the sea level, and therefore the volume of the continental ice sheets, have changed during the last 150,000 years, together with corresponding changes in the mean global surface temperature. This diagram suggests that after the climate optimum at 120,000 BP the ice made a limited initial advance over a period of about 10,000 yr and fluctuated about this value for another 30,000 yr before making a major surge and achieving about half its ultimate extent in about 8000 yr. Then, after a partial recession lasting about 30,000 yr, it continued to spread and attained its maximum coverage at 20,000 BP. During the whole of the glacial epoch the southern limit of the ice and snow cover advanced, on average, from 50°N to 40°N.

On this same diagram I have plotted the variations in annual insolation received northwards of 45°N due to changes in the earth's orbit using both Milankovitch's original calculations and the revised figures of Vernekar (1972). The amplitude of these variations is about 4×10^{18} cal/day or 1.5×10^{21} cal/yr, about 1% of the total radiation incident on this

area. The most important feature of Fig. 2 is the close correlation between the major advances and recessions of the ice and the 40,000 yr cycle of variations in solar insolation, the maxima of the ice cover being nearly coincident with the minima in the radiation received above 45°N.

Using Milankovitch's original data, I calculate that between 83,000 and 18,000 BP, when the ice began to recede, there was an overall deficiency of 4.5×10^{25} cal or 1000 cal for each gramme of ice formed. Venekar's revised data indicates a smaller deficiency of 2.5×10^{25} cal but still almost sufficient to compensate for the 677 cal/g liberated in the atmosphere during the formation of snow. Between 18,000 BP and the present, the total excess of insolation north of 45°N was 10^{25} cal (Milankovitch) or 1.4×10^{25} (Venekar). The excess between 16,000 and 6,000 BP, when about 90% of the ice, i.e. 40×10^6 km³, disappeared was 4.2×10^{24} cal (M) or 1.0×10^{25} cal (V) compared with the 3.2×10^{24} cal required to supply the latent heat of fusion. If, however, the *initial* melting of the ice was in fact as rapid as Fig. 2 suggests, it would be difficult to account for this in terms of the excess radiation available at the time, but initially the melting may have been aided by the release of heat stored in the oceans during the previous glacial epoch and by enhanced meridional transport through the atmosphere as suggested by Newell (1974). There is also the possibility that the fluctuations in the summer-time radiation (at 10,000 BP the excess northwards of 45°N was 10^{19} cal/day) had a proportionately greater cumulative effect on both the advance and the retreat of the ice than the annual variations. However, because of the losses and redistribution experienced by the solar radiation after entering the atmosphere and the effects of all the feedback processes, the formation and melting of the ice can hardly be explained solely in terms of long-term fluctuations in the radiation incident on top of the atmosphere. This will be a task for realistic numerical models of the atmosphere-ocean-cryosphere system incorporating seasonal effects, variable cloud and albedo, a fully interactive radiation scheme and interactions between the atmosphere and the oceans which will involve computation of transports of heat and salt by ocean currents.

The importance of meridional heat transport by the oceans for the advance and recession of global ice cover is emphasized in an interesting recent paper by Newell (1974). He proposes the existence of two quasi-stable regimes of atmospheric temperature and circulation corresponding to the alternating glacial and interglacial epochs and that these are characterized by two modes of partitioning of the poleward energy flux between the atmosphere and the oceans. In the present warm interglacial regime the oceanic transport across 30°N is about 3.3×10^{19} cal/day and that of the atmosphere about 5.1×10^{19} cal/day. Newell suggests that in a cold epoch the oceans carry less, and the atmosphere more, heat than at present, the total poleward transport being slightly less. During the advance of the glaciation the oceans gradually warm up and eventually cause the termination of the ice age; conversely, during the interglacial regime, the oceans cool slowly thereby arresting the retreat of the ice and leading to the next glaciation.

The mechanism is conceived as follows. As the ice recedes it exposes an increasing area of ocean surface water to cooling by radiation. This cold water, having a maximum density at 0°C, sinks and flows equatorwards as a slow, cold undercurrent. Being stably stratified, it inhibits both the upward transfer of heat from the ocean floor and downward transfer from the overlying surface layers. The sub-surface water therefore gradually cools and when it is brought to the surface, for example by upwelling off coasts with strong off-shore winds, the poleward transfer of heat by wind-driven ocean surface currents and circulations is reduced. This leads to further cooling at high latitudes and causes the melting and retreat of the ice to be arrested and eventually reversed. On the other hand, when the ice advances, it shields the underlying water from strong radiative cooling and reduces the source of the cold oceanic undercurrent so that the oceans gradually warm up and

increase their contribution to the poleward heat flux which is further enhanced by the stronger meridional temperature gradient imposed by extension of the ice. Again this provides a negative feedback mechanism for arresting the spread of the ice but it probably works only slowly so that the ocean surface and the overlying atmosphere remain cold and maintain extensive ice cover until the solar insolation increases.

Newell estimates the change in poleward energy flux necessary to cause a transition from a warm to a glacial epoch as follows. The difference between the volumes of land ice existing in the northern hemisphere 20,000 years ago and today is equivalent to about $47 \times 10^6 \text{ km}^3$ of water. Formation of this by the deposition of water vapour would release 677 cal/g or a total of $3.2 \times 10^{25} \text{ cal}$ in the polar atmosphere. Newell assumes that practically all of this was lost by radiation to space during a period of rapid glaciation lasting about 8000 yr and was not compensated by poleward transport through the atmosphere and oceans, the deficiency being 10^{19} cal/day , i.e. about one-third of the present-day oceanic transport across 30°N . We may note in passing that since only about half of the total volume of ice cited by Newell was formed in 8000 yr (see Fig. 2) a more accurate estimate for the deficiency would be $5 \times 10^{18} \text{ cal/day}$ in close agreement with the maximum deficiency in solar radiation plotted in Fig. 2.

Newell's mechanism is one interesting example of a possible internal feed-back process that might either amplify or damp the Milankovitch effect depending on their phase differences, but it seems unlikely that alternate slow cooling and warming of the deep ocean could, of itself, cause the alternate advance and recession of the ice sheets without external forcing. One must also consider other feedback mechanisms such as the effects of changing ice cover on albedo, evaporation, cloud amount and precipitation. It seems likely that increasing ice cover with its higher albedo will reduce further surface temperatures, evaporation and total precipitation, but perhaps increase the snowfall along the advancing ice edge thereby providing a positive feedback response for its further extension; on the other hand, the increased surface albedo will be at least partly offset by the reduced cloud cover. It is scarcely possible or meaningful to distinguish cause and effect in such a complex system of non-linear interactive processes, and impossible to predict the resultant effects without recourse to numerical models, the potentialities and limitations of which will be described in the next section.

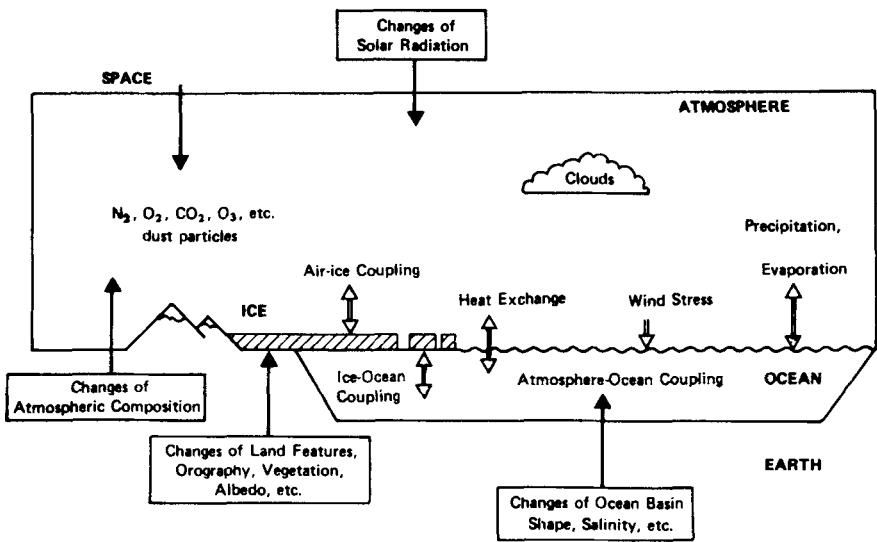


Figure 12. The physical processes and properties that govern the global climate and its changes. (From the US National Academy Report, 'Understanding Climatic Change', 1975.)

Meanwhile, the above simple calculations strongly indicate that the Milankovitch variations were of the right phase and order of magnitude to initiate the succession of major advances and recessions of the ice sheets although these were probably sustained through positive feedback, and ultimately terminated through negative feedback, by the mechanisms mentioned earlier. It is very unlikely, however, that these slow variations of insolation were responsible for the sharp climatic fluctuations of much shorter periods, e.g. temperature changes of several degrees over only a few hundred years, that occurred several times during the last 100,000 years. These were more likely to have resulted from internal instabilities within the climatic system triggered perhaps by the disintegration of the Antarctic ice dome as postulated by Wilson (1964, 1969). He suggests that recorded changes in sea level of 15–25 m could have been caused by the breaking away of huge masses of ice, of $6\text{--}10 \times 10^6 \text{ km}^3$ in volume which spread out to reach 50°S , increasing the average albedo of the southern hemisphere by perhaps 7% and reducing the average surface temperature by as much as 7°C .

6. THE NUMERICAL MODELLING OF CLIMATE

By far the most promising approach to the understanding and prediction of climatic change lies in the construction of physico-mathematical models of the atmosphere, oceans and cryosphere treated as a complex geophysical system and which allow for changes in atmospheric composition, soil moisture, vegetation, etc. Considerable success has been achieved in developing numerical models of the atmosphere for predicting the weather several days in advance and for the simulation of the global atmospheric circulation over periods of up to about one year. Such models are based on the dynamical equations of motion, the thermodynamic equations, the equations of radiative heat transfer, the equation of state of the atmospheric gas, a continuity equation for the water (vapour, liquid and solid) content of the atmosphere, and representation of the transfer of heat, moisture and momentum between the earth's surface and the atmosphere. The processes and mechanisms controlling the global climate are represented schematically in Figs. 12, 13.

One of the models developed in the Meteorological Office by Corby *et al.* (1972),

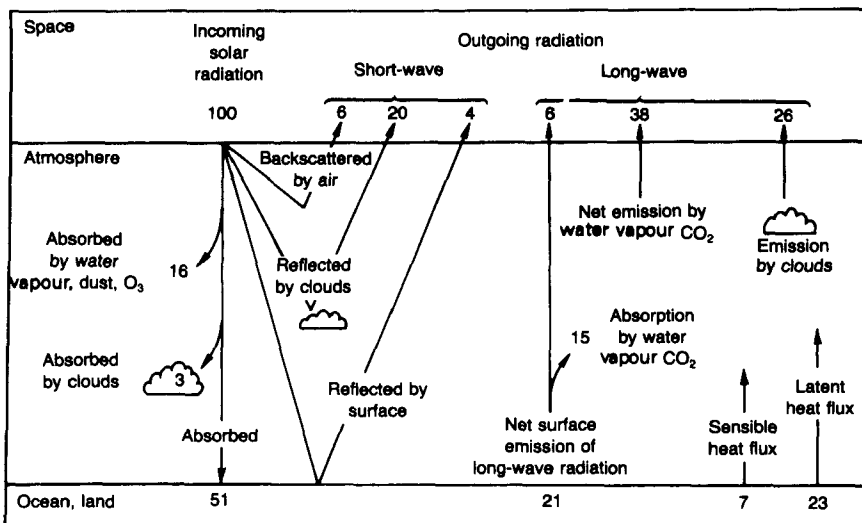
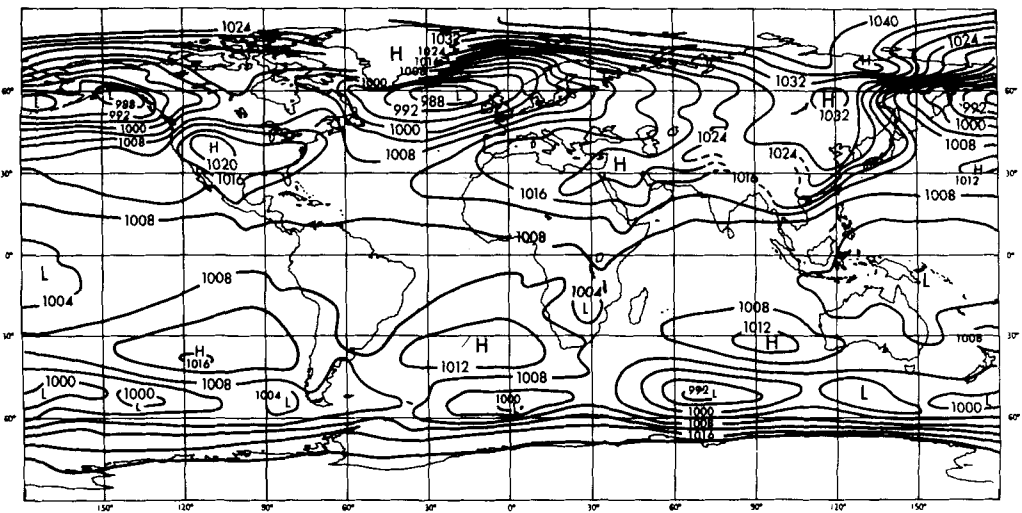


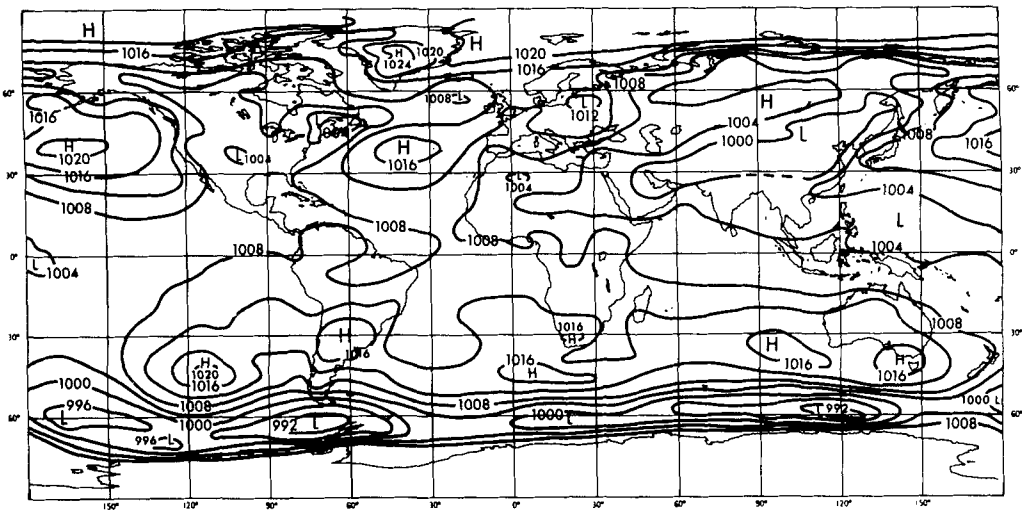
Figure 13. The radiation balance of the atmosphere–earth system showing the redistribution of the incoming solar radiation and the mechanisms by which these are effected. (From the US National Academy Report, ‘Understanding Climatic Change’, 1975.)

Gilchrist *et al.* (1973), represents the global atmosphere by five levels spanning the troposphere and lower stratosphere with 4600 grid points nearly evenly distributed at each level providing sufficient resolution to represent satisfactorily the thermal and dynamical structure of the atmosphere and details of the larger transient weather systems such as the depressions of middle latitudes. The earth's topography and the seasonal variations of the solar radiation entering the top of the atmosphere are prescribed. The temperatures of the sea surface are held at their seasonal average values but the temperatures of the land surface are determined by solving the relevant heat-transfer equations. The hydrological cycle is included but simplified to the extent that condensation is assumed to occur when the



PRESSURE M.S.L. JANUARY EX.385 MEAN D.61 TO 100

(a)



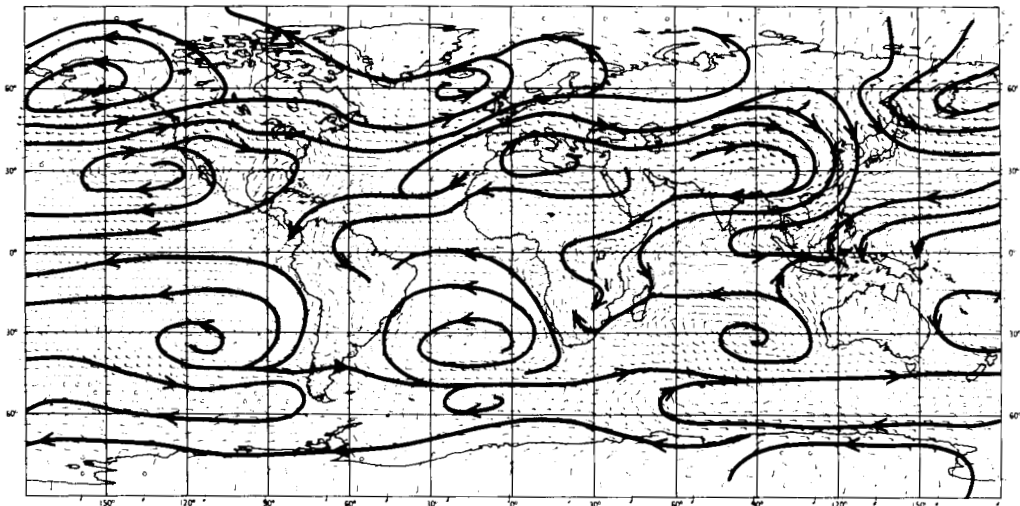
PRESSURE M.S.L. JULY EX.379 MEAN D.61 TO 100

(b)

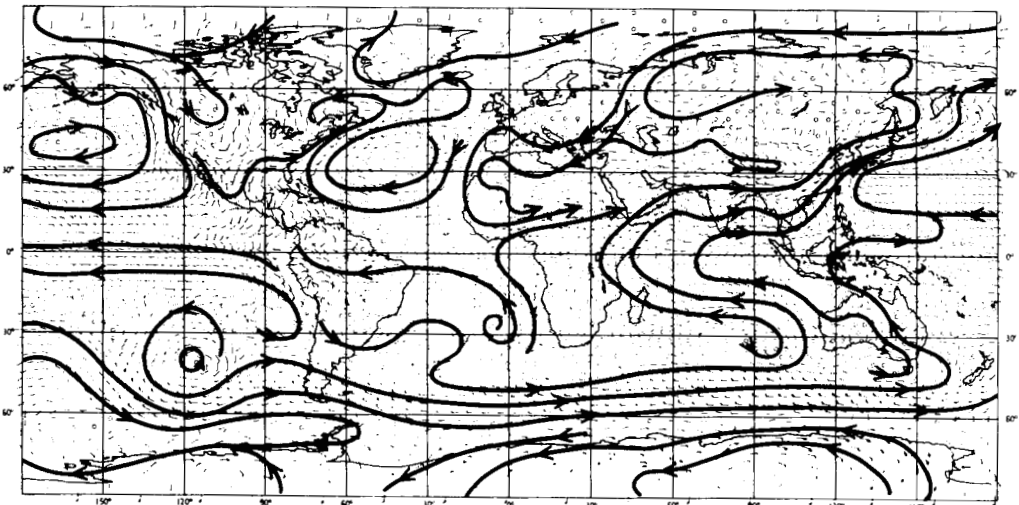
Figure 14. Simulation of the global distribution of mean sea-level pressure by the Meteorological Office 5-level model. (a) January, (b) July.

relative humidity of the air exceeds 100 per cent and the excess moisture is deemed to fall out as rain, allowance being made for evaporation if the precipitation falls through unsaturated air. The large-scale dynamical effects resulting from the release of latent heat of condensation are automatically computed but the effects of small-scale convective motions are allowed for only by rather crude statistical averaging.

A weakness of the model is that the type and amount of cloud (as distinct from precipitation) are not computed and so the important interactions between the clouds and the radiation field are not accurately represented. Because the oceans, being great storage reservoirs of heat, almost certainly exert a strong long-term control on the climate, it would be much more satisfactory to use a coupled atmosphere-ocean model that would



• 9 SIGMA LEVEL WINDS EX. 385 (JAN) D 61 TO 100 MEAN
(a)

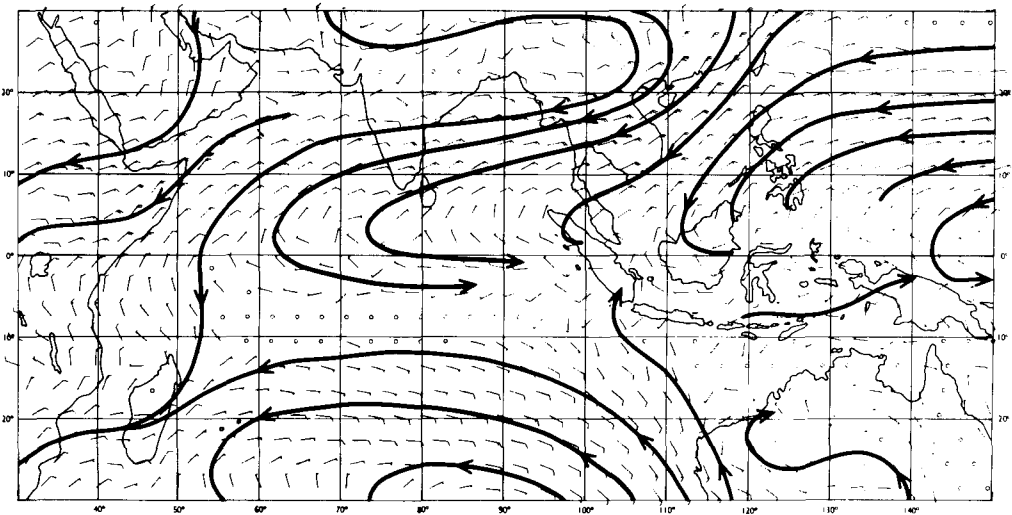


• 9 SIGMA LEVEL WINDS EX. 379 (JULY) D 61 TO 100 MEAN
(b)

Figure 15. Simulation of global winds systems for (a) January, (b) July.

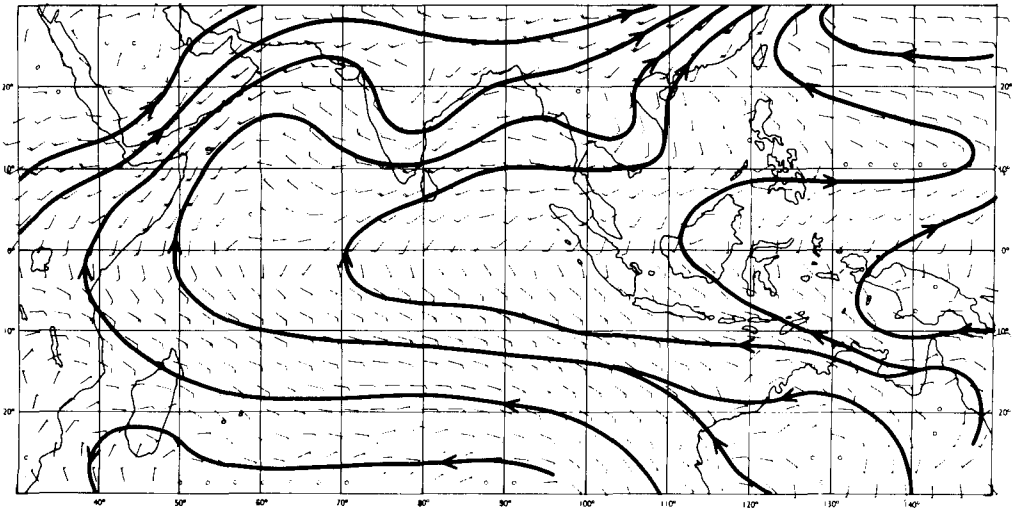
predict rather than prescribe the sea-surface temperatures and their changes, but such models are still in their infancy not least because ocean currents, temperatures, and salinities are not extensively and continually measured.

Nevertheless, despite these limitations, such atmospheric models successfully simulate the major features of the global circulation and of world climate. Starting from an initial observed situation, or from a motionless, isothermal, dry atmosphere of specified total mass and composition, the equations may be integrated forward in time steps of typically 10 min for periods of order one year without the model becoming dynamically or computationally unstable. Differential heating by the sun produces pressure gradients and consequently air motions. Gradually the winds strengthen until transient eddies closely similar



• 9 SIGMA LEVEL WINDS EX. 385 (JAN) D61 TO 100 MEAN

(a)



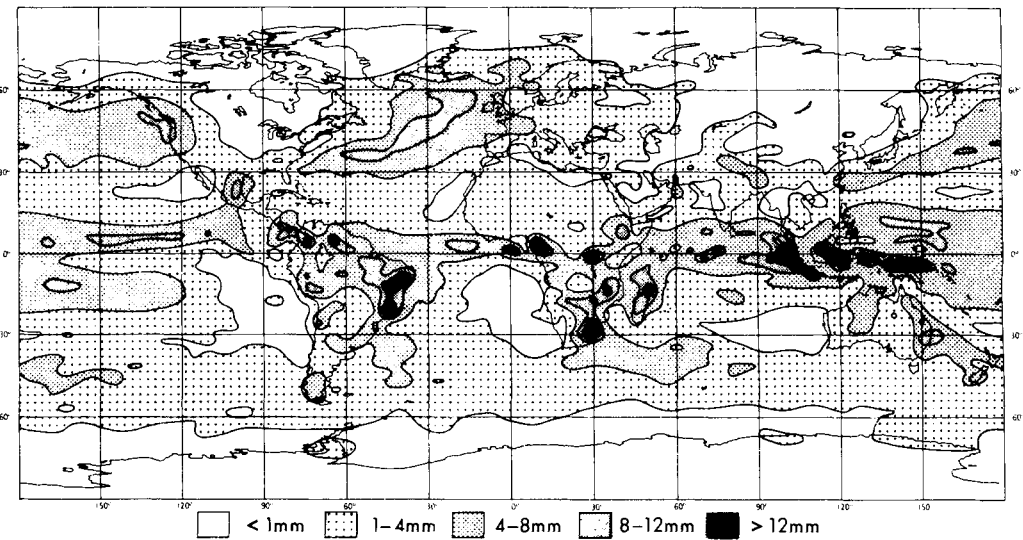
• 9 SIGMA LEVEL WINDS EX. 379 (JULY) D61 TO 100 MEAN

(b)

Figure 16. Simulation of the monsoon circulation for (a) January, (b) July.

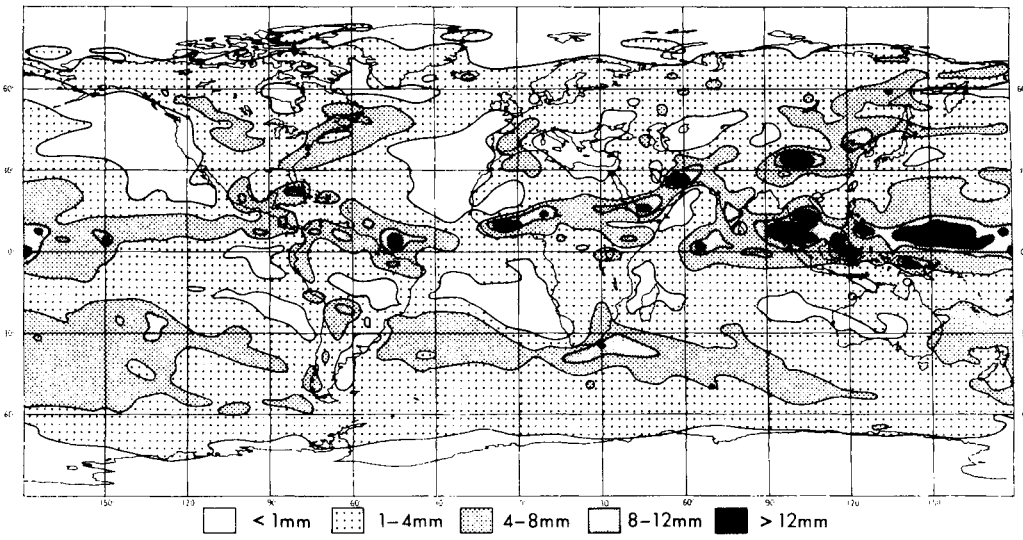
to the cyclones and anticyclones of the real atmosphere are generated. These redistribute heat and moisture and the model soon simulates quite well the seasonal distributions of temperature, pressure, wind and rainfall observed in the real atmosphere. After an integration extending over some 50 days, taking about 10 hours on the IBM 360/195 computer, the model reaches a state of statistical equilibrium to produce quantities which, when averaged over about a month, reproduce quite realistically the basic features of the earth's climate.

Figs. 14 (a) and (b) show the computed global distributions of surface pressure for January and July reproducing high-pressure cells over the continents (e.g. the Siberian anticyclone) and low-pressure systems (e.g. the Icelandic and Aleutian lows) over the oceans



RAINFALL EX.385 (JANUARY) D.61 TO 100 MEAN

(a)



RAINFALL EX.379 (JULY) D.61 TO 100 MEAN

(b)

Figure 17. Simulation of the global rainfall distribution for (a) January, (b) July.

in winter, the situation being reversed in the summer. The corresponding wind fields, represented by computed grid point values in Figs. 15 (a) and (b), reproduce the middle-latitude westerlies, the trade-winds and the doldrums together with the N.E. monsoon over Asia and East Africa in winter and replaced by the S.W. monsoon in summer. These monsoon circulations are shown in more detail in Figs. 16 (a) and (b). The chain of low-pressure centres off Antarctica, greatly intensified during the local winter, and the enhancement of the mid-latitude westerlies in winter, are also well simulated. The January and July distributions of global rainfall shown in Figs. 17 (a) and (b), while not accurate in detail, nevertheless reproduce the major features of the monsoon and tropical rainbelts and the desert and semi-arid regions quite well.

7. MODELLING POSSIBLE PERTURBATIONS IN CLIMATE

Numerical models of the type just described are now being used to determine how stable the climate is likely to be in response to conceivable changes in external or internal factors, either natural or man-made. One approach is to test the sensitivity of the statistics generated by the model to perturbations in a particular parameter. In order to yield consistent and meaningful results, the model should have a stable equilibrium and the magnitude of the perturbations should not be large enough to change the model climate from one equilibrium state to a markedly different one, but sufficiently large to distinguish significant effects from random fluctuations.

Fig. 13, which illustrates how the incoming solar radiation is absorbed, reflected and re-radiated within the atmosphere and at the earth's surface, suggests several ways in which this radiative balance could be modified and produce changes in the climate. These include variations in

1. the incoming solar radiation
2. the atmospheric composition, e.g. the carbon-dioxide, dust, ozone or water vapour content
3. the cloudiness
4. the albedo (reflectivity) of the earth's surface through changes in soil moisture, vegetation, ice and snow cover
5. the oceanic circulation leading to changes in sea-surface temperature and the transport of heat and moisture
6. the artificial release of large quantities of heat and moisture.

The effects of changing the 'solar constant' by a few per cent on a dynamical global circulation model of the atmosphere have been described in section 4. The effects of the Milankovitch variations have not yet been computed with such a model but this is now being attempted in the Meteorological Office using a 5-level global model incorporating a fully interactive cloud and radiation scheme.*

Since the total sensible heat generated by human activities is at present $<0.01\%$ of the total solar input to the atmosphere and earth, the global effects are likely to be well below a level that can be either calculated or detected.

The concentration of carbon-dioxide in the atmosphere has increased by about 10% during this century and is currently rising at about $\frac{1}{2}\%$ per annum due to the burning of fossil fuels. Since it strongly absorbs the longwave radiation emitted by the earth's surface, an increase in CO_2 should produce higher temperatures by the 'greenhouse effect'. Model calculations carried out by Manabe and Wetherald (1975), using a limited-area version of the model described on page 484, suggest that a 100% increase in CO_2 is likely to warm the

* See Addendum.

entire lower atmosphere by an average of about 3 degC, with the implication that a 10% increase would raise temperatures by only a few tenths of one degree. Within the next 50 years, fossil fuels will probably be largely exhausted and replaced by nuclear power stations emitting large quantities of waste heat into the atmosphere. It is considered likely that such stations will be concentrated in large floating off-shore parks using seawater for cooling, and that their heat output may be many times the local net influx of solar radiation. An experiment recently carried out on one of the Meteorological Office models suggests that such a concentrated emission of heat of 1.5×10^{14} W over an area of 4×10^5 km² or 375 W m^{-2} would cause significant changes in the circulation and increased rainfall extending over 1000 km downstream of the site.

The fact that temperatures in the northern hemisphere fell during the period 1940–1970, despite a steady increase in the concentration of carbon-dioxide, has been attributed to a simultaneous increase in the aerosol (dust) content of the atmosphere. Although the atmospheric turbidity even at remote sites increased markedly after 1963, the transmittance of direct solar radiation being decreased by 2%, this was almost certainly caused by a large volcanic explosion in Bali and now, with measurements almost back to pre-1963 values, there is little evidence that the dust content is increasing significantly. Following the Bali eruption, temperatures in the lower stratosphere rose by several degrees but there were no detectable effects at ground level. This observation is consistent with the results obtained from an experiment on the Meteorological Office global model which has 13 levels spanning both the troposphere and stratosphere. The insertion of a stratospheric layer of dust, sufficient to intercept 4% of the incoming solar radiation, produced local heating of up to 10 degC due to absorption of radiation by the dust but there were no discernible effects at ground level (see Fig. 18). This hardly supports the thesis that cooler epochs in the historical record may have been caused by volcanic eruptions.

The same model was used to investigate the likely effects of reducing the concentration of the stratospheric ozone layer by up to 50 per cent. This produced a cooling of up to

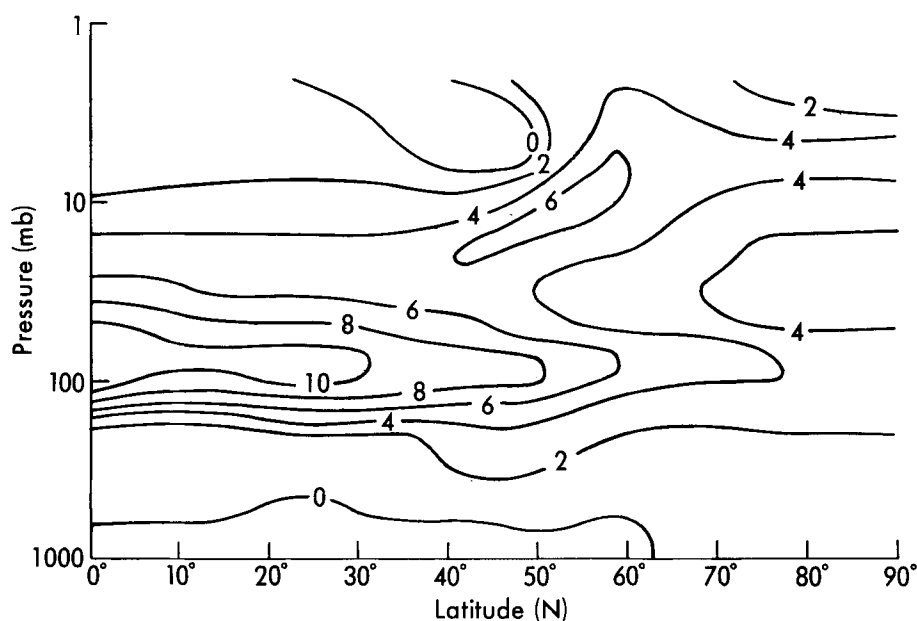


Figure 18. Changes in air temperature (K) predicted by the Meteorological Office 13-level global model as a consequence of introducing an artificial dust layer into the lower stratosphere sufficient to absorb 4% of the incoming solar radiation.

20 degC at 40 km altitude over the tropics but again insignificant changes in the lower atmosphere. Since investigations carried out in the Meteorological Office indicate that several hundred Concorde, each flying five hours per day, would not reduce the ozone by more than $\frac{1}{2}\%$ (US scientists have estimated about 1%), and such a small reduction could not be distinguished from the much larger natural fluctuations, supersonic transport aircraft are unlikely to have a significant effect on climate during this century.

Since the average albedo (reflectivity) of the earth's surface is 35% but varies from about 8% for dark green vegetation to about 80% for freshly fallen snow, widespread changes in vegetation, ice/snow cover or soil moisture could produce a significant change in albedo and hence in climate. On the whole man's activities tend to raise the albedo and so reduce the fraction of solar radiation available to warm the land surface. Model calculations indicate that a 1% change in global albedo would cause a change of about 1 degC in mean global temperature. In an experiment with the Meteorological Office 5-level model, the Arctic ice sheet was assumed to melt and be replaced by a water surface held at 0°C. The major effects of removing the ice, apart from the obvious one of warming the polar troposphere, were to weaken the intense polar anticyclone, to diminish the strength of the middle-latitude westerlies and their associated depressions, and to produce a significant cooling of up to 8 degC in middle latitudes especially over the United States, eastern Siberia and western Europe.

At this stage the results of such model experiments should be regarded as indicative rather than definitive partly because the models still have serious deficiencies in that they do not properly treat the interactions between cloud and the radiation field nor the changes in the surface layers of the oceans and their interaction with the atmosphere.

The influence of anomalies of sea-surface temperature on the atmospheric circulation has been investigated in some detail by Rowntree (1976), using the Meteorological Office 5-level model. During the winter of 1962/3, the coldest in Britain for 250 years, a large area of the eastern tropical Atlantic Ocean was up to 2.5 degC warmer than normal. When this anomaly was inserted into the model it produced an area of low surface pressure centred west of the Bay of Biscay and, in one experiment, an extensive area of high pressure centred just east of Greenland. The modified circulation resulted in a strong easterly flow over the British Isles reminiscent of that which produced the very cold winter of 1962/3.

8. PREDICTION OF CLIMATIC VARIATIONS

If we define climate as the statistical properties of the atmosphere averaged over an ensemble of all the states occurring within a long but finite time span during which the atmosphere adjusts to slowly changing external conditions, climatic prediction is concerned with determining how these statistics will change as the time span advances and the external conditions change.

Although a model may well *simulate* quite well the essential features of the existing (or any other quasi-equilibrium) climate, it does not necessarily follow that it will be able to *predict* a unique sequence of climatic events from a given perturbation of this equilibrium state. The degree of predictability will depend upon the degree to which these events are uniquely determined by the boundary conditions and external forcing factors and to what extent they are influenced by internal fluctuations within the climatic system itself.

Protagonists of dynamical models tend to regard climatic change as an initial boundary-value problem in fluid dynamics which, for fixed external boundary conditions, is capable of a deterministic solution, i.e. an equilibrium climate with statistics independent of the initial state. A change of climate is expected to result only from a change in the external or boundary conditions but, even so, a useful prediction could be made only if the

external changes themselves were predictable and the range and accuracy of the prediction would depend upon the amplitude and amplification of the external stimulus relative to the noise generated within the climatic system. Even the most advanced of current dynamical models, though heavily constrained by fixed or only slowly changing boundary conditions, are not completely deterministic in the sense that the same model starting from rather different initial states is apt to produce rather different sets of climatological statistics, the differences of which may mask the effects being sought. Consequently the models though sufficiently realistic and robust to *indicate* the likely climatic effects of large changes in input parameters are not yet able to *predict* the quantitative effects of relatively small perturbations. To what extent this indeterminacy is a consequence of the deficiency of the model and to what extent it is an inherent property of a system involving non-linear interactions between many processes operating on a wide range of time scales will be settled more convincingly by extensive experimentation than by philosophical arguments. But even if dynamical models should predict a unique series of responses to a given set of external stimuli, there is still the possibility that the *real* climatic system may be able to exist in two (or more) quasi-stable modes and change rather abruptly from one such state to another. The transitions between glacial and interglacial epochs shown in Fig. 1 are suggestive of such changes. Whether these are externally induced, as suggested in section 5, or whether they can arise spontaneously from internal reconfiguration of the climatic system, will again require running computer models for very long periods and the attendant risk that they will become unstable for computational reasons.

Meanwhile, attempts are being made to circumvent the vast amount of computation required by explicit dynamical models, by developing statistical-dynamical models in which the properties and structure of individual large eddies (e.g. cyclones) would not be computed explicitly but their transport properties would be parameterized in terms of the large-scale features of the wind and temperature fields. However there are considerable difficulties in this approach and many results of detailed global circulation models would be required to carry out the parameterization. Moreover, because the cyclones, which are so important in converting potential energy into kinetic energy, tend to be suppressed in simple models with poor resolution, it is difficult to envisage a realistic climate model that does not handle them properly. But it may be that low resolution models with clever parameterization of the unresolved scales, will determine the mean fields and the important eddy transports with sufficient accuracy to indicate how the climate will respond to identifiable long-term variations in external forcing and boundary conditions even if they are unable to predict smaller and more subtle fluctuations that, nevertheless, may have serious economic consequences.

For the immediate future priority should probably be given to the use of models to test the sensitivity of the atmosphere's response to changes in individual parameters, to elucidate the underlying physical mechanisms, and to distinguish likely changes in atmospheric behaviour from the idiosyncrasies of particular models.

ADDENDUM

INCORPORATION OF MILANKOVITCH EFFECT INTO A GCM MODEL

(The results of these computations were presented at the meeting by Mr A. Gilchrist)

To provide an indication of the effects of variations in the earth's orbital parameters on temperatures during northern summer, two integrations have been run using a global version of the Meteorological Office 5-layer general circulation model incorporating the changes made by Francis (1975) and an interactive radiation scheme with three cloud layers.

The two integrations were for 60 days, representing May and June, and used different orbital parameters representing the present day and 10,000 BP as given in the table.

	Present day 10,000 BP	
Eccentricity	·0167	·0190
Longitude of perihelion	249°	102°
Obliquity	23·45°	24·22°

The effect of the first two changes was that, at the northern summer solstice, the earth was receiving about 7% more solar radiation than at present with net annual radiation virtually unchanged; the change in obliquity increased the annual radiation at the poles by about 3%, with small decreases in the tropics. To avoid the model's thermal structure being determined by the initial sea surface temperatures, the ocean was given a depth of 2 metres; this no doubt allowed a larger atmospheric response than would be obtained with a proper ocean model interacting with the atmospheric model over many years. Averaged for days 51–60 (late June) the run for 10,000 BP was warmer than the present day run throughout the atmosphere with differences ranging from 6 degC over the Arctic to 4 degC at 30°N, 2 degC at 30°S and 0 degC at the South Pole.

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